Iterated integrals

1 Iterated integrals

1.1 Motivation: Picard's method for differential equations

Let $A(t)$ be a $n \times n$ matrix of continuos functions on an open $U \subset \mathbb{R}$ and consider the initial-value problem

$$\begin{cases} \dot{\underline{X}}(t) = A(t)\underline{X}(t) \\ \underline{X}(t_0) = \underline{X}_0 \end{cases} . \tag{1}$$

This system is equivalent to the Volterra integral equation

$$\underline{X}(t) = \int_{t_0}^{t} A(s)\underline{X}(s)\,\mathrm{d}s. \tag{2}$$

By interpreting this equation as fixed-point problem, one can approximate the solution with an iterative method. Consider the sequence $\{\underline{X}_0(t), \underline{X}_1(t) \dots, \underline{X}_n(t), \dots\}$, where

$$\underline{X}_n(t) = \underline{X}_{n-1} + \int_{t_0}^{t} A(s)\underline{X}_{n-1}(s)\mathrm{d}s. \tag{3}$$

and $\underline{X}_0(t) \equiv \underline{X}_0$. If a solution exists, the limit of the sequence is the solution to the differential equation. To understand the structure of $\underline{X}_n$ let us consider for example the case $n = 2$

$$\underline{X}_2(t) = \underline{X}_1(t_0) + \int_{t_0}^{t} A(s)\underline{X}_1(s)\,\mathrm{d}s = \tag{4}$$

$$= \underline{X}_0 + \int_{t_0}^{t} \mathrm{d}s\, A(s)\underline{X}_0(s) + \int_{t_0}^{t} \mathrm{d}s \int_{t_0}^{s} \mathrm{d}s' A(s)A(s')\,\underline{X}_0. \tag{5}$$

Iterating, we see that

$$\underline{X}_n(t) = \underline{X}_0 \left(1 + \sum_{m=1}^{n} \int_{\Delta_m} A(s_1) \dots A(s_m)\mathrm{d}s_m \dots \mathrm{d}s_1\right), \tag{6}$$

where Δ_m is the m-dimensional simplex defined by $t_0 \leq s_1 \leq \cdots \leq s_m$. If the matrix A commutes with itself at different t's, the sum can be explicitly resummed. Observe that, in this case, the integrand is symmetric upon exchange of the argument of integrations, and, therefore, the integral domain can be extended to the hypercube $[t_0, t]^m$ with a factor of $1/m!$ to cancel the overcounting of the new regions. The integrals the reduce to the power m of a single integral and

$$\underline{X}(t) = \exp\left(\int_{t_0}^{t} A(s)\,\mathrm{d}s\right)\underline{X}_0. \tag{7}$$

Example. Consider the one-dimensional differential equation $\dot{x} = tx$. By separation of variables, one can verify that $x(t) = x_0 e^{\frac{t^2}{2}}$. By iterated integration one can compute the first terms of the sequence $\left\{x_0, x_0\left(1 + \frac{t^2}{2}, x_0\right), x_0\left(1 + \frac{t^2}{2} + \frac{t^4}{2\cdot 3}\right)\right\}$ and see the series of the exponential arising.

1.2 Definition of iterated integrals

Let M be a smooth manifold over the field $\mathbb{K}$ ($\mathbb{R}$ or $\mathbb{C}$). Let $\gamma : [0,1] \to M$ a piecewise smooth path and $\omega_1, \dots, \omega_n$ $\mathbb{K}$-valued 1-forms on M. Let us first recall that the ordinary line integral of a 1-form is

$$\int_\gamma \omega = \int_{[0,1]} \gamma^*(\omega) = \int_0^1 f(t)\,\mathrm{d}t, \tag{8}$$

where $\gamma^*(\omega)$ denotes the pull-back. Recall that this integral is parametrization independent.

Definition. *The **iterated integral** of $\omega_1, \dots, \omega_n$ along γ is*

$$I_\gamma(\omega_1, \dots \omega_n) = \int_\gamma \omega_1 \dots \omega_n = \int_{0 \leq t_1 \leq \dots \leq t_n} f_1(t_1) \dots f_n(t_n)\, dt_n, \tag{9}$$

where the case $n = 0$ is defined to be one and we refer to iterated integrals as linear combinations of such.

Iterated integrals satisfy a number or properties.

Proposition.

- *Iterated integrals do not depend on the parametrization of γ;*
- *denoting by $\gamma^{-1}(t) = \gamma(1-t)$ the path reversal,*

$$I_{\gamma^{-1}}(\omega_1, \dots, \omega_n) = (-1)^n I_\gamma(\omega_n, \dots, \omega_1); \tag{10}$$

- *given two consecutive paths α and β,*

$$I_{\alpha\beta}(\omega_1, \dots, \omega_n) = \sum_{i=0}^{n} I_\alpha(\omega_1, \dots, \omega_i) I_\beta(\omega_{i+1}, \dots, \omega_n); \tag{11}$$

- *iterated integrals satisfy the shuffle product algebra*

$$I_\gamma(\omega_1, \dots, \omega_r) I_\gamma(\omega_{r+1}, \dots, \omega_{r+s}) = \sum_{\sigma \in Shuffle(r,s)} I_\gamma(\omega_{\sigma(1)}, \dots, \omega_{\sigma(r+s)}). \tag{12}$$

1.3 Homotopy invariance

Fixed the endpoints of γ and the 1-forms, some integrals are **homotopy functionals**, that is, they depend only homotopy class the curve belongs to. The case of the line integral is well known.

Lemma. $\int_\gamma \omega$ is an homotopy functional $\Leftrightarrow \omega$ is closed

Therefore, in one real or complex dimension, all line integrals are trivially homotopy functionals.

Example. Let us compute $I_\gamma(\frac{dz}{z})$, where $\gamma(0) = 1$ and $\gamma(1) = i$. The 1-form $\frac{dz}{z}$ is smooth in $\mathbb{C}/\{0\}$. First, consider the linear path $\gamma_1(t) = (1-t) + it$

$$I_{\gamma_1}(dz/z) = \int_{\gamma_1} \frac{dz}{z} = \int_0^1 \frac{i-1}{1+(i-1)t}\, dt = \mathrm{Log}(i) = i\frac{\pi}{2}. \tag{13}$$

Now consider the anti-clockwise circular arc joining the points $\gamma(t) = e^{i\frac{\pi}{2}}$

$$I_{\gamma_1}(dz/z) = \int_{\gamma_2} \frac{dz}{z} = \int_0^1 \frac{i\pi}{2}\, dt = i\frac{\pi}{2}. \tag{14}$$

Finally, consider clockwise circular arc $\gamma_3(t) = e^{-i\frac{3\pi}{2}t}$. From the previous line it is evident that $I_{\gamma_3}(\frac{dz}{z}) = -i\frac{3\pi}{2}$. The first two integrals are the same because the paths are homotopy invariant. The final path is not since it goes around the puncture of the complex plane.

Example. Let $M = \mathbb{R}^2$ and $\gamma_{r,s}(t) = (t^r, t^s)$ a family of homotopical equivalent paths. Consider the 1-forms $\omega_1 = \mathrm{d}x$ and $\omega_2 = \mathrm{d}y$, the iterated integrals along the γ's are[1]

$$\int_\gamma \omega_1\omega_2 = \int_0^1 dt_1 \int_{t_1}^1 dt_2\, r t_1^{r-1} s t_2^{s-1} = \frac{s}{r+s}, \tag{15}$$

hence $I_\bullet(\omega_1, \omega_2)$ is not a homotopy functional.

To determine wether an iterated integral is a homotopy functional, we first need to define the **bar construction**. Let M be a connected manifold and X a collection of smooth 1-forms in M[2]. Let us denote by $[\omega_1|\dots|\omega_n]$ the elements of $X^{\otimes n}$, called **words of depth** n. Consider now the derivation

$$D([\omega_1|\dots|\omega_n]) := \sum_{i=1}^{n} [\omega_1|\dots|\mathrm{d}\omega_i|\dots|\omega_N] + \sum_{i=1}^{n-1} [\omega_1|\dots|\omega_i \wedge \omega_{i+1}|\dots|\omega_n]. \tag{16}$$

Definition. *A word of depth n $\xi \in X^{\otimes n}$ is **integral** if $\xi \in B(M) = Ker(D)$.*

$B(M)$ is a commutative algebra in the shuffle product naturally filtrated by the depth n. Notice that $B_1(M) \cong \mathbb{K} \oplus H^1_{dR}(M, \mathbb{K})$. To every integrable word we associate an iterated integral

$$[\omega_1|\dots|\omega_n] \mapsto I_\gamma(\omega_1, \dots, \omega_n) = \int_\gamma \omega_1 \dots \omega_n. \tag{17}$$

Theorem. The integration map defines an isomorphism between the set of integrable words $B_{\leq n}(M)$ and the homotopy invariant interated integrals of lenth $\leq n$.

From this theorem we then recover the case of the line integral. Notice also that this implies that in one real or complex dimension all line integrals are homotopy functionals.

[1] Note the correct extrema of integration.
[2] A *model* of M, whatever it means.

1.4 The case of the punctured complex plane

Let us now consider the complex plane with a finite number of punctures. All the iterated integrals on such manifold are homotopy invariant.

1.4.1 $M = \mathbb{C}$

First consider $M = \mathbb{C}$ and $\omega = \mathrm{d}z$. Rather than considering fixed endpoints, we fix $\gamma(0)$ and let $\gamma(1) = z$ be a variable. In such way iterated integrals define functions in each distinct homotopy class, that is to say **multivalued-functions**. Since the fundamental group of $\mathbb{C}$ is trivial, the functions we will find are actually-single valued. Using homotopy invariance we can fix $\gamma(0) = 0$ and take the linear path $\gamma(t) = tz$. Then the line integral is

$$\int_\gamma \omega = \int_0^1 z\, dt = z. \tag{18}$$

The iterated integral of length n is

$$\int_\gamma \omega \dots \omega = \int_{0\le t_1 \le \dots \le t_n} z^n \,\mathrm{d}t_1 \dots \mathrm{d}t_2 = z^n \mathrm{Vol}(\Delta_n) = \frac{z_n}{n!}. \tag{19}$$

Therefore, the iterated integrals on $\mathbb{C}$ give rise to polynomials.

1.4.2 $M = \mathbb{C}/\{0\}$

Let us now consider $\mathbb{C}/\{0\}$. The form we iterate on is $\omega = \frac{dz}{z}$. We could fix $\gamma(0) = 0$ and introduce a regularization, but this slightly complicates the matter. Let us take then $\gamma(0) = 1$ and $\gamma(1) = z$ and the line integral over γ. Since the fundamental group is $\mathbb{Z}$, there will be a family of $\mathbb{Z}$ different values depending on which homotopy class is γ taken from. Let us first pick the line path $\gamma(t) = tz + (1-t)$.

$$I_\gamma(\omega) = \int_\gamma \frac{dz}{z} = \int_0^1 \frac{z-1}{tz+1-t} dt. \tag{20}$$

By splitting the real and imaginary part of the integrand by taking $z = a + ib$, we find

$$I_\gamma(\omega) = \log\sqrt{a^2+b^2} + i \arctan\frac{b}{a}. \tag{21}$$

We recognize in the RHS the definition of the principal value of the $\mathrm{Log}\, z = \log|z| + i\mathrm{Arg}(z)$. If we now take $\gamma_k' = \gamma \circ \gamma_k$, where γ_k is a loop winding around $k \in \mathbb{Z}$ times around $z = 0$ having $z = 1$ as base point, we get

$$I_{\gamma_k'}(\omega) = I_\gamma(\omega) + I_{\gamma_k}(\omega) = I_\gamma(\omega) + i2k\pi, \tag{22}$$

which are the other branches of the complex logarithm. As expected, the iterated integral defines a multi-valued function by taking paths from different homotpy classes. Indeed, one could take this to be the *definition* of the analytical continuation of the logarithm.
Finally, iterated integrals of ω are just powers of the logarithm

$$\int_\gamma \omega \dots \omega = \frac{1}{n!}\log^n(z), \tag{23}$$

where the equality should be intended in a multi-valued sense.
We could also take γ starting from 0 and introduce a regularization prescription. In this case it is also necessary to introduce a different notion of homotpy by looking at the value of $\gamma'(0)$.

1.5 The doubly punctured complex plane

We now move to the first truly non trivial case, $M = \mathbb{C}/\{0,1\}$. The 1-forms we consider are $\omega_0 = \mathrm{d}z/z$ and $\omega_1 = \mathrm{d}z/(1-z)$. Let us pick as base point $\gamma(0) = 0$. The line integrals on each give $\log(z)$ (properly regularized) and $-\log(1-z)$, in a multi-valued sense. The generic iterated integral will then be integrals on $\log's$ with those kernels. We will see that these will give rise to special functions called generalized polylogarithms. Let us first introduce the subfamily of classical polylogarithms and show that they are produced by a class of iterated integrals.

1.5.1 Classical polylogarithms

Let us define on the complex disc $|z| < 1$ the classical polylogarithms as the absolutely convergent series

$$\mathrm{Li}_n(z) = \sum_{k=1}^{\infty} \frac{z^k}{k^n}. \tag{24}$$

The case $n = 1$ is

$$\mathrm{Li}_1(z) = \sum_{k=1}^{\infty} \frac{z^k}{k} = -\log(1-z). \tag{25}$$

Notice that

$$\frac{d}{dz}\mathrm{Li}_n(z) = \sum_{k=1}^{\infty} \frac{z^{k-1}}{k^{n-1}} = \frac{1}{z}\mathrm{Li}_{n-1}(z). \tag{26}$$

Since $\mathrm{Li}_n(0) = 1$,

$$\mathrm{Li}_n(z) = \int_\gamma \mathrm{Li}_{n-1}(w)\frac{dw}{w} = \int_\gamma \mathrm{Li}_{n-1}(w)\omega_0, \tag{27}$$

where γ is a smooth path from 0 to z in M. For appropriate γ and z in the unit disk, we retrieve the series definition, while for generic paths and z's we get a multi-valued function on M. By iterating the recursive relation, we find

$$\mathrm{Li}_n(z) = \int_\gamma -\log(1-w)\omega_0\cdots\omega_0 = \int_\gamma \omega_1\omega_0\cdots\omega_0. \tag{28}$$

By choosing the linear path, we get

$$\mathrm{Li}_n(z) = \int_{0\le t_1\le\cdots\le t_n} \frac{z dt_1}{1-zt_1}\frac{dt_2}{t_2}\cdots\frac{dt_n}{t_n}. \tag{29}$$

Notice that even though the path starts in 0, the integral is convergent when ω_0 does not appear as the first form. From this integral one can also recover the series expansion by applying the geometric series on the first kernel and integrating each term separately.

Example. Let us study in a bit more detail the **dilogarithm**

$$\mathrm{Li}_2(z) = \int_\gamma \omega_1\omega_0. \tag{30}$$

where $\gamma(0) = 0$ and $\gamma(1) = z$. Let us take first the linear path $\gamma(t) = tz$. The iterated integral can be written in two ways

$$\mathrm{Li}_2(z) = \int_0^1 \frac{z dt_1}{1-zt_1}\int_{t_1}^1 \frac{dt_2}{t_2} = \int_0^1 \frac{dt_2}{t_2}\int_0^{t_2}\frac{z dt_1}{1-zt_1}. \tag{31}$$

The first equality shows explicitly the $\mathrm{Li}_2(z)$ has no singularity in $z = 0$. From the second we find

$$\mathrm{Li}_2(z) = -\int_0^1 dt\,\frac{\log(1-zt)}{t}. \tag{32}$$

If $z = x \in \mathbb{R}$ the change of variables $xt = u$ gives the usual real representation of the dilogarithm

$$\mathrm{Li}_2(x) = -\int_0^x \frac{\log(1-u)}{u}. \tag{33}$$

1.5.2 Generalized polylogarithms

We have seen that strings of $\omega_1\omega_0\cdots\omega_0$ generate classical polylogarithms. Now, fixing ω_1 to be the first form, a generic insertion of r ω_1's can be represented as $\omega_1\omega_0^{n_1-1}\omega_1\cdots\omega_0^{n_r-1}$. We recover $\mathrm{Li}_n(z)$ with $r = 1$ and $n_r = n$. The functions that arise in this case are called generalized polylogarithms, defined in the unit disk by the series

$$\mathrm{Li}_{n_1\ldots n_r}(z) = \sum_{1\le k_1<\cdots<k_r} \frac{z^{k_r}}{k_1^{n_1}\ldots k_r^{n_r}}. \tag{34}$$

Again, we see that

$$\frac{d}{dz}\mathrm{Li}_{n_1\cdots n_r}(z) = \begin{cases} \frac{1}{z}\mathrm{Li}_{n_1\cdots n_r-1}(z) \text{ if } n_r > 1 \\ \frac{1}{1-z}\mathrm{Li}_{n_1\cdots n_{r-1}}(z) \text{ if } n_r = 1 \end{cases}. \tag{35}$$

While the first case is the same as beforem the second require a bit more work. We work out explicitly an example

$$\mathrm{Li}_{3,1}(z) = \frac{d}{dz}\sum_{k_1=1}^{\infty}\sum_{k_2=k_1+1}^{\infty}\frac{z^{k_2}}{k_1^3 k_2} = \tag{36}$$

$$= \sum_{k_1=1}^{\infty}\frac{1}{k_1^3}\sum_{k_2=k_1+1}^{\infty} z^{k_2-1} = \sum_{k_1=1}^{\infty}\frac{1}{k_1^3}\left(\frac{1}{1-z} - \frac{1-z^{k_1}}{1-z}\right) = \frac{1}{1-z}\sum_{k_1=1}^{\infty}\frac{z^{k_1}}{k_1^3} = \frac{1}{1-z}\mathrm{Li}_3(z). \tag{37}$$

From this recursion we read off the desired iterated integral which furnishes the polylogs with their analytical continuation

$$\mathrm{Li}_{n_1\ldots n_r}(z) = \int_\gamma \omega_1\omega_0^{n_1-1}\omega_1\ldots\omega_1\omega_0^{n_r-1}. \tag{38}$$

1.5.3 Regularization

Finally, we can take care of strings starting with ω_0 by applying the multiplicative property of polylogs. Consider for example

$$\int_\gamma \omega_0 \int_\gamma \omega_1 = \int_\gamma \omega_0\omega_1 + \int_\gamma \omega_1\omega_0,$$

the LHS is well-defined making use of the mentioned regularization prescription in the case of the singly-punctured complex plane. The second term of the RHS has been computed in the previous sections. We then defined the first term to be

$$\int_\gamma \omega_0\omega_1 := \int \omega_0\omega_1 - \int_\gamma \omega_0 + \int_\gamma \omega_1 = \mathrm{Li}_2(z) + \log(z)\log(1-z). \tag{39}$$

1.5.4 Alphabets and words

Let us formalize the notion of strings of form we have mentioned previously. We call the set $X = \{x_0, x_1\}$ *alphabet* of two *letters*. We denote the free monoid generated by X with X^* and its elements we call *words*. Linear combinations of words with rational coefficients are denoted by $\mathbb{Q}\langle x_0, x_1\rangle$ and they form a vector space. Furtherly, they form a commutative algebra with the shuffle product

$$x_{i_1}\dots x_{i_r} \sqcup\!\sqcup x_{i_{r+1}}\dots x_{i_{r+s}} = \sum_{\sigma\in\Sigma(r,s)} x_{\sigma_1}\dots x_\sigma(r+s).$$

Now to each word we define an iterated integral based on the discoveries of the previous sectoin. If $w \in X^*x_1$ can be written as $w = x_0^{n_r-1}x_1\dots x_1x_0^{n_1-1}x_1$ and we define

$$\mathrm{Li}_w(z) = \int_\gamma \omega_!\omega_0^{n_1-1}\cdots = \mathrm{Li}_{n_1\dots n_r}(z). \tag{40}$$

If $\omega \in X^*$ we extend the map using the previous regularization and by imposing, in addition

$$\mathrm{Li}_{x_0^n}(z) = \int_\gamma \omega_0\dots\omega_9 = \frac{1}{n!}\log^n(z). \tag{41}$$

This can always be done by observing that every word w can be written uniquely as

$$w = \sum_{i=0}^{k} x_0^i \sqcup\!\sqcup w_i \tag{42}$$

where $w_i \in X^*x_1$.

1.5.5 Summary

We have a complete description of the iterated integrals on $M = \mathbb{C}/\{0,1\}$ by the following isomorphism

$$(\mathbb{C}\langle X\rangle, \sqcup\!\sqcup) \to \{\text{Homotopy-invariant iterated integrals on } M\}. \tag{43}$$

Any iterated integral on M is a linear combination of $\mathrm{Li}_w(z)$, which are linearly independent functions. This can be shown to be equivalent to the statement that polylogs are stable under multiplication and taking primitives.

1.6 Multivariate case

Polylogarithms are generalized in higher dimensions by extending the variety to the moduli space genus 0 Riemann surfaces with $n+3$ marked points

$$M = \mathfrak{M}_{0,n+3} = \{(t_1,\dots,t_n) \in \mathbb{C}^n : t_i \neq t_j, t_i \neq 0,1\}. \tag{44}$$

The $n = 1$ case is the same as the doubly-punctured complex plane. For any n, one has the family of $n(n-3)/2$ differential forms

$$\Omega = \left\{ \frac{dt_i - dt_j}{t_i - t_j}, \frac{dt_i}{t_i}, \frac{dt_i}{1-t_i}] \right\}. \tag{45}$$

Since $\mathrm{d}\omega_i = 0$ for all $\omega_i \in \Omega$, the integrable words satisfy

$$[\omega_1|\dots|\omega_i \wedge \omega_{i+1}|\dots|\omega_N] = 0$$

for all i. The special functions which arise in this context are multivariate generalized polylogarithms, called **Gonchrov polylogarithms**

$$\mathrm{Li}_{n_1,\dots,n_r}(x_1,\dots,x_r) = \sum_{0<k_1<\cdots<k_r} \frac{x_1^{k_1}\dots x_r^{k_r}}{k_1^{n_1}\dots k_r^{n_r}}, \tag{46}$$

and the iterated integrals extend these to multi-valued function on the whole moduli space. Similarly as before we have

Theorem. Every homotopy invariant iterated integral on $\mathfrak{M}_{0,n+3}$ can be expressed as sum of products of $\log(t_i)$ and multiple polylogarithms.

One of the main differences with the univariate case is that this expansion now is not unique. A base for the iterated integral was found by Francis Brown.

2 Monodromy

2.1 Fuchsian differential equation

Let us consider again the differential equation (1) and promote to the complex plane (or, moore generically, to a genus 0 Rieamann surface)

$$\begin{cases} d\underline{X}(z) = A(z)\underline{X}(z) \\ \underline{X}(z_0) = \underline{X}_0 \end{cases}, \tag{47}$$

where now $A(z)$ is a square matrix of 1-forms. We say this differential equation is **Fuchsian** if $(A(z))_{ij}$ contains only simple poles. Let the poles at finite distance be $\{a_1, \dots a_k\}$. This equivalent to say that $A(z) - \frac{A_k}{z-a_j}dz$ contains no poles in a_j. The solution to this differential equation is given by the complex Dyson series realized by iterated integrals

$$\underline{X}(z) = \left(\sum_{m=0}^{\infty} \int_\gamma A \dots A \right) \underline{X}_0, \tag{48}$$

where $\gamma(0) = \underline{X}_0$ and $\gamma(1) = z$. The fundamental group of $X = \mathbb{C}/\{a_1, \dots, a_k\}$ is the abelian group freely generated by the loops γ_i around each a_i passing through a fixed base point z_0. Consider the solution 48 evaluated at $z = z_0$ along the trivial path in $\gamma_0 \in \pi_1(X, z_0)$. All the other paths can be obtained by composing γ_0 with any other $\gamma_i \in \pi_1(X, z_0)$ any number of times. We denote the solution computed along $\gamma = \gamma_0 \circ \gamma_i$ with $\mathcal{M}_i\underline{X}(z)$. The operator $\mathcal{M}_i$ represents the continuation of any function along the path γ_i.

Proposition. *Consider the complete set of initial conditions encoded by the identity matrix $\mathbb{I}_{n\times n}$. Let V be the matrix of the solutions to the n Cauchy problems. We have $\mathcal{M}_i V = V \exp(2\pi i A_i)$.*

The matrix $M_i = \exp(2\pi i A_i)$ is called **monodromy matrix** and all such matrices form the **monodromy group**. Given any matrix M, finding a linear system of ODE's that admit M as a monodromy matrix is the statement of the 21st Hilbert's problem (solved).

Example. Consider the system of ODE's

$$d\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} = \begin{pmatrix} 0 & \frac{dz}{1-z} \\ 0 & 0 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}. \tag{49}$$

We have

$$V = \begin{pmatrix} 1 & -\log(1-z) \\ 0 & 1 \end{pmatrix}. \tag{50}$$

The monodromy matrix associated to $z = 1$ is

$$M_1 = \exp\left(\begin{pmatrix} 0 & 2\pi i \\ 0 & 0 \end{pmatrix} \right) = \begin{pmatrix} 1 & 2\pi i \\ 0 & 1 \end{pmatrix}, \tag{51}$$

and, indeed,

$$\begin{pmatrix} \mathcal{M}_1(1) & \mathcal{M}_1(-\log(1-z)) \\ \mathcal{M}_1(0) & \mathcal{M}_1(1) \end{pmatrix} = \begin{pmatrix} 1 & -\log(1-z) \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2\pi i \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -\log(1-z) + 2\pi i \\ 0 & 1 \end{pmatrix}, \tag{52}$$

as expected.
The example can be extended by including all powers of log by replacing the 2×2 matrix in (49) with a $(p+1)\times(p+1)$ matrix with $\frac{dz}{1-z}$ on the elements just above the diagonals. For example, we find that

$$\mathcal{M}_1 \log^2(z) = \log^2(1-z) - 2i\pi \log(1-z) - 2\pi^2. \tag{53}$$

Notice that in the previous example the solution to the differential equation could be computed exactly because the matrix is unipotent so the iteration stops in a finite number of steps. Functions arising from such systems are said to have **unipotent monodromy**.

Example. Consider the square root functions $X(z) = \sqrt{z}$. since it satisfies the differential equation

$$dX = \frac{dz}{2z} X, \tag{54}$$

$\mathcal{M}_\prime = -1$. Therefore the monodromy group is $\mathbb{Z}_2$.

2.2 Monodromy of polylogarithms

Following the previous section we compute the monodromy of polylogarithms in a low-dimensional example first, and then of all generalized polylogarithms through the Knizhnik–Zamolodchikov (KZ) equations.
Consider the linear system with

$$A(z) = \begin{pmatrix} 0 & \frac{dz}{z} & 0 \\ 0 & 0 & \frac{dz}{1-z} \\ 0 & 0 & 0 \end{pmatrix}. \tag{55}$$

The evolution operator is

$$U(z, z_0) = \mathbb{I} + \int_\gamma \begin{pmatrix} 0 & \frac{dz}{z} & 0 \\ 0 & 0 & \frac{dz}{1-z} \\ 0 & 0 & 0 \end{pmatrix} + \int_\gamma \begin{pmatrix} 0 & 0 & \frac{dz}{z}\frac{dz}{1-z} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \tag{56}$$

the solution

$$V(z) = \begin{pmatrix} 1 & -\log(1-z) & \mathrm{Li}_2(z) \\ 0 & 1 & \mathrm{Li}_1(z) \\ 0 & 0 & 1 \end{pmatrix}. \tag{57}$$

Now we have two monodromies since we have two distinct singular 1-forms, their explicit representation is

$$M_0 = \begin{pmatrix} 1 & 2\pi i & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad M_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 2\pi i \\ 0 & 0 & 1 \end{pmatrix}, \tag{58}$$

and we have

$$\begin{cases} \mathcal{M}_0 \mathrm{Li}_2(z) = \mathrm{Li}_2(z) \\ \mathcal{M}_1 \mathrm{Li}_2(z) = \mathrm{Li}_2(z) + 2\pi i \log(z) \end{cases}. \tag{59}$$

Now let us extend this result by defining a generating functional for the polylogarithms. First, let x_0 and x_1 abstract letters generating the X^* words. The univariate **Knizhnik-Zamoldchikov (KZ) equation** is

$$\frac{d}{dz} L(z) = \left(\frac{x_0}{z} + \frac{x_1}{1-z} \right) L(z), \tag{60}$$

where $L(z)$ should be regarded as a $\mathbb{C}\langle\langle X \rangle\rangle$, that is, a linear combination of words with complex coefficients. Using Picard's method we see that

$$U = 1 + \int_\gamma \left(\frac{dz}{z} x_0 + \frac{dz}{1-z} x_1 \right) + \int_\gamma \left(\frac{dz}{z}\frac{dz}{z} x_0^2 + \frac{dz}{z}\frac{dz}{1-z} x_0 x_1 + \frac{dz}{1-z}\frac{dz}{z} x_1 x_0 + \frac{dz}{1-z}\frac{dz}{1-z} x_1^2 \right). \tag{61}$$

whence we see that

Proposition. *A solution to the KZ equation is*

$$L(z) = \sum_{w \in X^*} Li_w(z) w. \tag{62}$$

Another way to see this is to write $L(z)$ as

$$L(z) = 1 + \sum_{w \in X^*} \mathrm{Li}_{x_0 w}(z) x_0 w + \sum_{w \in X^*} \mathrm{Li}_{x_1 w}(z) x_1 w. \tag{63}$$

It easy then to verify explicitly that this expression satisfies the equation using that

$$\frac{d}{dz} \mathrm{Li}_{x_0 w}(z) = \frac{1}{z} \mathrm{Li}_w(z), \tag{64}$$

$$\frac{d}{dz} \mathrm{Li}_{x_1 w}(z) = \frac{1}{1-z} \mathrm{Li}_w(z). \tag{65}$$

The first few terms of $L(z)$ are

$$L(z) = 1 + \mathrm{Li}_{x_0}(z) x_0 + \mathrm{Li}_{x_1}(z) x_1 + \mathrm{Li}_{x_0^2}(z) x_0^2 + \mathrm{Li}_{x_1 x_0}(z) x_1 x_0 + \mathrm{Li}_{x_0 x_1}(z) x_0 x_1 + \mathrm{Li}_{x_1^2}(z) x_1^2 + \cdots = \tag{66}$$

$$= 1 + \log(x) x_0 - \log(1-x) x_1 + \tag{67}$$

$$+ \frac{1}{2} \log^2(z) x_0^2 + \mathrm{Li}_2(z) x_1 x_0 + (\mathrm{Li}_2(z) + \log(z) \log(1-z)) x_0 x_1 + \frac{1}{2} \log^2(1-z) x_1^2 + \ldots. \tag{68}$$

Since we are interested in monodromy, let us consider the singularities of $L(z)$. By looking at the first terms, we have that of the coefficients are regular around $z = 0$ except $\log^n(z)$. In general

$$\begin{cases} \lim_{z \to 0} \mathrm{Li}_w(z) = 0 \text{ if } w \neq x_0^n \\ \lim_{z \to 0} \mathrm{Li}_{x_0^n}(z) = \frac{1}{n!} \log^n(z) \end{cases}. \tag{69}$$

Then,

$$\lim_{z\to 0} L(z) = 1 + \log(z)x_0 + \frac{1}{2}\log^2(z)x_0^2 + \cdots = e^{x_0\log(x)}. \tag{70}$$

By $x_0 \leftrightarrow x_1$ symmetry, there exist another solution to (60) that has the same singularity structure in $z = 1$

$$\lim_{z\to 1} L^1(z) = e^{-x_1\log(1-x)}. \tag{71}$$

Now, the previous statementss on the monodromy can be applied to this abstract case here

$$\begin{cases} \mathcal{M}_0 L(z) = L(z)e^{2\pi i x_0} \\ \mathcal{M}_1 L^1(z) = L^1(z)e^{2\pi i x_1} \end{cases}. \tag{72}$$

From the first relation we can work out the monodromies of the polylogs around $z = 0$ by comparing coefficients. From (69), we know that the only non-trivial result is for the powers of $\log(z)$. In particular we have

$$\mathcal{M}_0 L(z) = \mathcal{M}_0\left(1 + \sum_{w\in X^*} \mathrm{Li}_{x_0 w}(z)x_0 w + \sum_{w\in X^*} \mathrm{Li}_{x_1 w}(z)x_1 w\right) = \tag{73}$$

$$= \left(1 + \sum_{w\in X^*} \mathrm{Li}_{x_0 w}(z)x_0 w + \sum_{w\in X^*} \mathrm{Li}_{x_1 w}(z)x_1 w\right)(1 + 2\pi i x_0 + (2\pi i)^2 x_0^2 + \dots), \tag{74}$$

let us compare the terms with coefficient x_0^n

$$\mathcal{M}_0 \mathrm{Li}_{x_0^n}(z) = \mathcal{M}_0\left(\frac{\log^n(z)}{n!}\right) = \sum_{k=0}^{n} \frac{\log^k(z)}{k!}(2\pi i)^{n-k}. \tag{75}$$

Now, in order to use the second equation to compute the monodromy of polylogs around $z = 1$ we need to relate the solutions $L(z)$ and $L^1(z)$. Let us define the **Drinfeld associator**

$$\Phi(z) = (L^1(z))^{-1}L(z). \tag{76}$$

By differentiating each side of $L^1(z)\Phi(z) = L(z)$ we find that $d\Phi(z) = 0$, that, it actually series of words with *constant* coefficients. Since it is constant, we can evaluate at any point, and, in particular,

$$\Phi = \lim_{z->1^-} \exp(\log(1-z)x_1)L(z). \tag{77}$$

Therefore, we can regard Φ as a regularization of $L(z)$ around 1 (which contains powers of $\log(1-z)$). In particular

Proposition.

$$\Phi = \sum_{w\in X^*} \zeta(w)w, \tag{78}$$

where $\zeta(w)$ is a real valued function such that

- $\zeta(x_0) = \zeta(x_1) = 0$*;*
- $\zeta(x_0^{n_r-1}x_1\dots x_0^{n_1-1}x_1) = \zeta(n_1,\dots,n_r) = \sum_{0<k_1<\cdots<k_r} \frac{1}{k_1^{n_1}\dots k_r^{n_r}}$ *(called* ***multiple zeta values****);*
- $\zeta(w)\zeta(w') = \zeta(w \sqcup\!\sqcup w')$.

One can verify this by expanding (77). All the terms containing powers $\log(1-z)$ are cancelled against the terms arising from the exponential and the multiple zeta values simply arise by noticing that $\mathrm{Li}_{x_0^{n_r-1}x_1\dots x_0^{n_1-1}x_1}(1) = \zeta(n_1,\dots,n_r)$. For example, the terms up to depth 2, are

$$\Phi = 1 + \zeta(2)x_0x_1 - \zeta(2)x_1x_0 + \dots \tag{79}$$

Finally, using the Drinfeld associator, it is easy to see that

$$\mathcal{M}_1 L(z) = L(z)\Phi^{-1}e^{2\pi i}\Phi. \tag{80}$$

Example. As an application let us use this formula to find the monodromy of classical polylogarithms. From both sides of the equation we need to extract the coefficient of $x_0^{n-1}x_1$. The RHS greatly simplifies by considering that we cannot take terms whose last letter is not a single x_1. Therefore[3]

$$\mathcal{M}_1(\mathrm{Li}_n(z)x_0^{n-1}x_1) = \left(1 + \sum_w L_{x_0 w}(z)x_0 w\right)(\Phi^{-1}\Phi + \Phi^{-1}2\pi i x_1\Phi + \dots). \tag{81}$$

[3]In the following, for the equality to make sense, we are implicitly throwing away all terms proportional to words different than $x_0^{n-1}x_1$.

Since Φ comes after x_1, only the empty word remains. Similarly, Φ^{-1} should contains only powers of x_0, but $\zeta(x_0^k) = 0$, so again only the empy word contributes. Then

$$\mathcal{M}_1(\mathrm{Li}_n(z)x_0^{n-1}x_1) = \left(1 + \sum_w L_{x_0 w}(z)x_0 w\right)(1 + 2\pi i x_1) = L_{x_0^{n-1}x_1}(z)x_0^{n-1}x_1 + 2\pi i L_{x_0^{n-1}}(z)x_0^{n-1}x_1, \tag{82}$$

and

$$\mathcal{M}_1\mathrm{Li}_n(z) = \mathrm{Li}_n(z) + \frac{2\pi i}{(n-1)!}\log^{n-1}(z). \tag{83}$$